GarageBand VR Drums

Week 12 Testing Plan | Interactive Prototype 2b

GarageBand VR Drums is an interactive virtual drumming experience that allows users to select a musician, play a virtual drum kit, record a short performance, replay or retake it, and save their recording for a virtual concert performance. Instead of interacting with a traditional music application, users are placed in a virtual performance environment where they can play the drums using Meta Quest controllers. The prototype aims to make the drumming and recording experience engaging, immersive, and easy to understand through a guided, game-like interaction flow.

Following the Week 9 usability evaluation, I made several improvements based on the issues identified during testing. These include fixing the cymbal sound and improving drum feedback, redesigning the Help panel to improve readability, and adding labels that appear when users hover over buttons on the recording panel. The Week 12 testing will evaluate whether these changes have improved the overall usability of the experience.

Testing Objective

The Week 9 evaluation identified several usability issues, particularly with drum feedback, Help panel readability, and understanding certain recording controls. Participants also experienced difficulty identifying the physical buttons on the Meta Quest controllers. Based on these findings, I refined the prototype to address these issues.

This test aims to evaluate whether the improvements made after Week 9 have addressed the previously identified usability issues. In particular, I want to examine whether the improved drum feedback provides a more responsive playing experience, whether the redesigned Help panel is easier to read and understand, and whether the hover labels help participants distinguish between recording controls such as Retake, Playback, and Save.

Participants will receive a brief introduction to the Meta Quest controllers, and I may provide instructions about which controller buttons to use during testing when necessary. This will allow me to focus on evaluating the revised features rather than participants' familiarity with VR hardware. The test will also identify whether any previous issues remain or new usability problems have appeared following the changes. The results will be evaluated through observed errors or hesitation, participants' understanding of the recording controls and system feedback, and their post-test feedback.

Testing Methodologies

This testing plan uses moderated task-based usability testing, direct observation, and think-aloud to evaluate the revised Unity prototype using a Meta Quest headset and controllers.

Before testing, participants will be introduced to the physical Meta Quest controller buttons, including the trigger, A, B, X, and Y buttons. During the tasks, I may provide instructions about which controller buttons to use if participants are unfamiliar with the hardware or have difficulty progressing. However, the functions of the recording panel buttons, such as Retake, Playback, and Save, will not be explained until participants have attempted to identify them using the hover labels. Participants will be encouraged to think aloud by describing what they expect to happen, whether the drum feedback feels responsive, and anything

they find confusing. Their actions, comments, and any unexpected behaviour will be observed to determine whether the revised features have improved the experience.

The results will also be compared with the Week 9 findings to evaluate whether previously identified usability issues have been addressed, while considering any assistance provided during testing.

Prototype Description/Requirements

The revised prototype maintains the guided seated VR drumming experience. Users can select a musician, start recording, play the drums using VR controllers, stop recording, retake or replay their performance, and save it for a virtual concert performance. Users can also pause or resume the performance and choose to delete their recording and record again. Following the Week 9 evaluation, I improved the cymbal sound and drum feedback to provide a more consistent playing experience. I also redesigned the Help panel to improve readability and added explanatory labels that appear when users hover over buttons on the recording panel. These improvements allow me to evaluate whether the revised interface and feedback have addressed the usability issues identified in Week 9.

Data Collection Method

During the testing process, I will use the Voice Memos app to audio-record each session with the participant's consent. This will allow me to focus on observing how participants interact with the prototype without having to write down every comment during testing, as doing so may cause me to miss important behaviours or reactions. After testing, I will transcribe the recordings to review participant comments and feedback in more detail. I will also use the Notes app to record any interesting or unexpected behaviours observed during the session, such as hesitation, incorrect button selections, difficulties using the Help panel, or issues with drum feedback. These notes will complement the audio recordings, particularly for behaviours that may not be captured through audio alone. After testing, I will analyse the transcripts and observation notes together to identify recurring usability issues, evaluate whether the improvements made after Week 9 have addressed previous problems, and determine any further changes needed. Participants will be labelled P1, P2, and so on rather than by name.

Testing Setup

Before each testing session, the prototype will be opened at the starting screen, and the Meta Quest headset and controllers will be checked to ensure they are functioning correctly. This includes checking musician selection, drum audio and hit detection, recording, playback, retake, save, concert performance, Help panel, and hover labels. Any previous recording state will be reset so that each participant begins with the same experience. The Voice Memos and Notes apps will also be prepared for data collection. Before starting the tasks, participants will receive a brief introduction to the physical Meta Quest controller buttons, including the trigger, A, B, X, and Y buttons. During testing, I may provide instructions about which controller inputs to use when necessary, while allowing participants to identify the recording panel controls themselves before offering assistance. The testing will aim to involve at least five participants.

Testing Process

1. Welcome and Consent (Approximately 1 minute)

"Thank you for testing my GarageBand VR Drums prototype. During this session, I will give you a series of tasks to complete using the prototype. Please try to complete each task on your own first. If you have any questions, feel free to ask. I may also help you with the VR controller buttons if needed. Please think aloud

where possible and tell me what you are looking for, what you expect to happen, or anything you find confusing. Is it okay if I record this testing session for observation and evaluation purposes?"

Audio recording will only begin after the participant has given their consent.

2. Meta Quest Controller Introduction (Approximately 1 minute)

"Before we begin, I'll briefly introduce you to the Meta Quest controllers. This is the trigger under your index finger, which you can use to select buttons in the experience. These are the A and B buttons on the right controller, and the X and Y buttons on the left controller. During the testing, I'll guide you through the tasks and let you know which controller buttons to press when needed. You can also ask me if you have any questions."

Participants will be introduced to the physical controller buttons before beginning the tasks. I may remind them where the buttons are located during testing if they have difficulty finding them.

3. Task Completion and Observation (Approximately 3-4 minutes)

1. **Task 1 - Help Panel Readability (Approximately 20 seconds):** "Before starting the experience, press X to open the Help panel and take a moment to read the instructions." Observe whether the Help panel text is readable within the headset, including text size, contrast, and layout. Note any difficulty reading or understanding the instructions.
2. **Task 2 - Start and Choose a Musician (Approximately 20 seconds):** ""Now, please start the experience and choose a musician you would like to play as. You can use the trigger to select the buttons." Observe whether the participant can follow the onboarding instructions, select a musician, and continue. Note any confusion or unexpected behaviour.
3. **Task 3 - Start Recording (Approximately 5 seconds):** "Now, let's start recording your drum performance. Press A to begin recording." Observe whether the recording starts correctly and the participant recognises that recording has begun.
4. **Task 4 - Play Drums (Approximately 30 seconds):** "Now, try playing the drums using the controllers. You can try different drums, including the cymbals." Observe whether the drums respond consistently when hit, whether the sounds play at the expected time, and whether the participant experiences any difficulties or unexpected responses.
5. **Task 5 - Stop Recording (Approximately 5 seconds):** "Okay, now that you've finished your performance, press A again to stop recording." Observe whether the recording stops correctly and the participant recognises that the recording has ended.
6. **Task 6 - Retake and Playback (Approximately 30 seconds):** "Imagine you're not satisfied with your first recording and want to try again. Look at the recording panel and try to find the option that lets you record again. You can hover over the buttons to see their labels and use the trigger to select one. Once you've finished your new recording, try to find the option to listen to your performance." Observe whether the participant notices and understands the hover labels, can distinguish between Retake and Playback, and experiences any confusion when selecting the controls.
7. **Task 7 - Save and Concert Performance (Approximately 10 seconds):** "Now that you're happy with your recording, try to find the option to save your performance. You can hover over the buttons to see their labels." Observe whether the participant understands the Save label and the transition to the concert performance.
8. **Task 8 - Pause and Resume Concert Performance (Approximately 15 seconds):** "Now that your recording is playing as a concert performance, please try pausing it. Once it has paused, try resuming the performance." Observe whether the participant can identify the Pause and Resume

controls, whether the performance responds correctly, and whether the participant understands the changes in playback state.

9. **Task 9 - Delete and Rerecord (Approximately 30 seconds):** "Imagine you want to replace your saved performance with a new recording. Please find the option to delete your current recording and start again." Observe whether the participant can identify the Delete and Rerecord option, understands what will happen to their saved performance, and can return to the recording process without confusion.

4. Post-Test Questions (Approximately 2-3 minutes)

After completing the tasks, participants will be asked the following questions:

1. "Was there any point during the experience where you felt confused or unsure what to do next?"
2. "Were the instructions in the Help panel clear and easy to read? Was there anything you found difficult to understand?"
3. "Did you notice the labels that appeared when you hovered over the recording buttons? Did they help you understand what each button does?"
4. "How did the drums feel when you were playing? Did the sounds respond as you expected, particularly the cymbals?"
5. "What did you think about the overall experience? Was there anything you particularly liked or disliked?"
6. "If you could change or improve one thing about the experience, what would it be?"

Total estimated testing time: Approximately 7-8 minutes per participant.